

4 All living organisms are made of cells.

Some cells are specialised for their function.

- (a) One specialised cell is the sperm cell.

Draw a mammalian sperm cell showing the acrosome, the nucleus and a mitochondrion.

Label the acrosome, nucleus and mitochondrion.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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- (b) A human egg cell is another example of a specialised cell.

- (i) A human egg cell has a radius of $50\text{ }\mu\text{m}$.

Calculate the **volume** of this human egg cell.

Use the formula:

$$V = \frac{4}{3}\pi r^3$$

(1)

Answer μm^3

- (ii) The surface area of this egg cell is $31\,416\text{ }\mu\text{m}^2$.

Give the surface area to volume ratio of this egg cell.

(1)

Answer : 1



- (iii) Explain **three** ways in which a mammalian egg cell is specialised for its functions.

(3)

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- (c) A fertilised egg cell develops into a morula and then a blastocyst.

Which of the following are correct statements?

1. A blastocyst contains pluripotent stem cells.
2. A blastocyst is a hollow ball of cells whereas a morula is not.
3. Both of these structures contain cells which divide by meiosis.

(1)

- ☐ **A** 1 and 2 only
- ☐ **B** 1 and 3 only
- ☐ **C** 2 and 3 only
- ☐ **D** 1, 2 and 3

(Total for Question 4 = 9 marks)

